# *Airport Management System*

***Prepared by:*** *Ajla Alcani*

***Course:*** *Software Modeling and Design*

***Document Overview:***

*This document presents the state machine models for the Airport Management System (AMS). It illustrates the exact states, events, and transitions of the system's main objects.*

***Objective :***

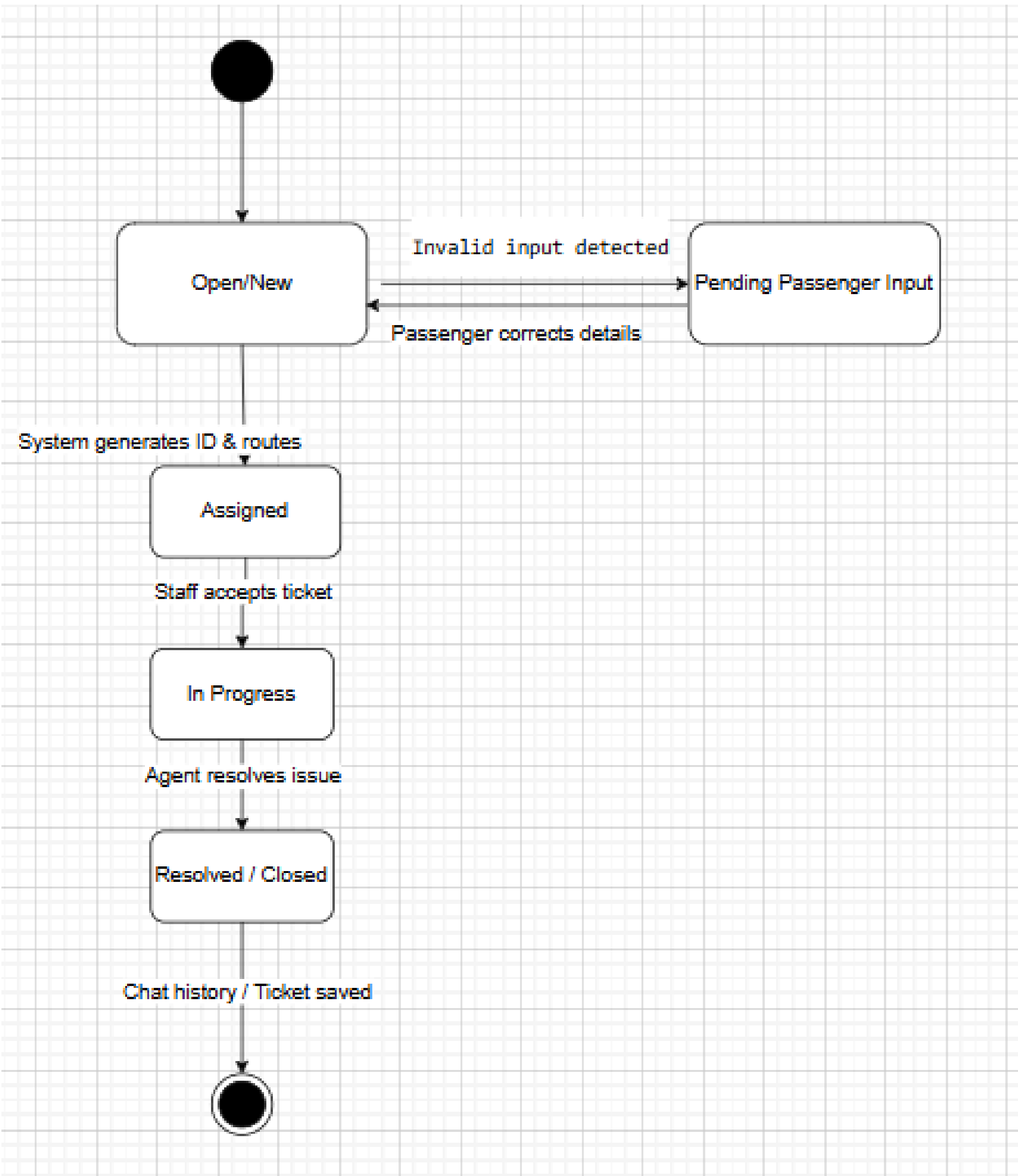
*The goal is to clearly map the behavioral workflows and object lifecycles to ensure the system design logically fulfills its operational requirements and serves as a foundation for implementation.*

## *State Machine Diagrams*

*This section defines the dynamic behavior and state transitions for the core system objects managed within the Customer Support and Human Resource modules.*

### *1.1 Support Ticket Lifecycle (Customer Support Module)*

- *Core Purpose: To model the exact states a support ticket passes through from initial creation by a passenger to final resolution by a support agent.*
- *Mapped Requirements: Ensures a ticket is successfully created, tracked, managed, and that the query is either resolved or escalated without data loss.*

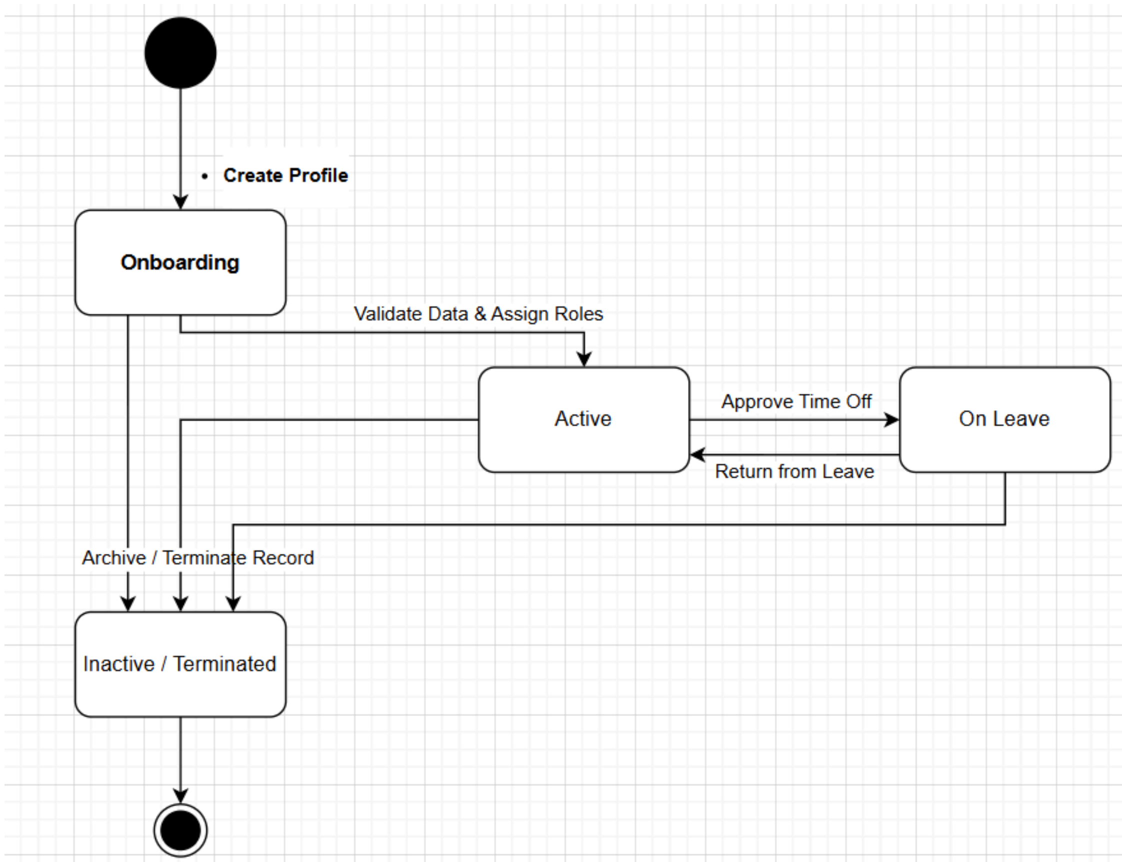


***State Descriptions:***

- ***Open/New:*** *The ticket has been submitted by the passenger but not yet reviewed.*
- ***Pending Passenger Input:*** *The system or agent has detected invalid/missing data and is waiting for the passenger to provide corrections.*
- ***Assigned:*** *The system has generated a unique ID and routed the ticket to an available Customer Support Agent.*
- ***In Progress:*** *The support staff has accepted the ticket and is actively working on resolving the passenger's issue.*
- ***Resolved / Closed:*** *The issue has been fixed, the passenger is notified, and the chat history/ticket data is saved to the system archives.*

### *1.2 Employee Record Lifecycle (Human Resource Module)*

- ***Core Purpose:*** *To model the operational states of an airport staff member's profile within the system, from initial hiring to departure.*
- ***Mapped Requirements:*** *UC_53 (Manage Airport Staff Records). Ensures HR strictly controls staff access, scheduling eligibility, and payroll status based on their current employment state.*




## ***State Descriptions & Transitions:***

- ***Onboarding (Initial State):*** ** Trigger: HR staff creates a new employee profile.*
    - *Description: The record is created, but the employee cannot yet be scheduled for shifts or processed for payroll. Awaiting document validation.*
- ***Active:*** ** Trigger: HR validates data and assigns staff roles.*
    - *Description: The employee is fully registered. They can now use the system to check in/out (UC_54) and are actively included in the payroll process (UC_55).*
- ***On Leave (Suspended State):*** ** Trigger: HR updates the schedule to reflect approved time off (vacation, medical leave).*
    - *Description: The employee remains in the system but is temporarily blocked from being assigned to daily operational shifts.*
- ***Inactive / Terminated (Final State):*** ** Trigger: HR deletes or archives the employee record.*
    - *Description: The employee no longer has system access. Historical payroll and attendance data are retained for auditing, but no new actions can be taken.*

***Airport Management System***

***Prepared by:*** *Ameo Lulaj*

***Course:*** *Software Modeling and Design*

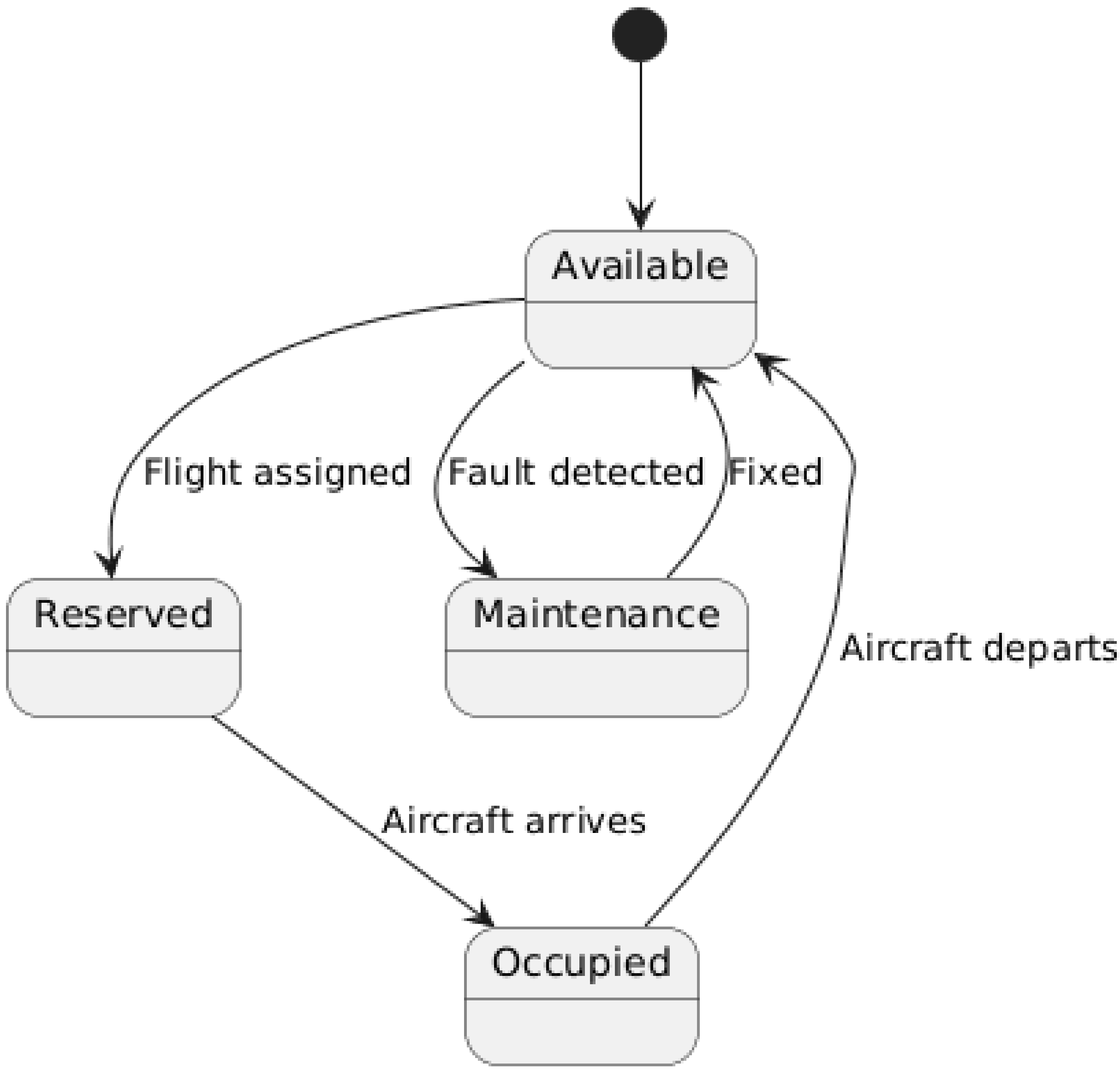
***Document Overview:***

*This document presents the state machine models for the Airport Management System (AMS). It illustrates the exact states, events, and transitions of the system's main objects.*

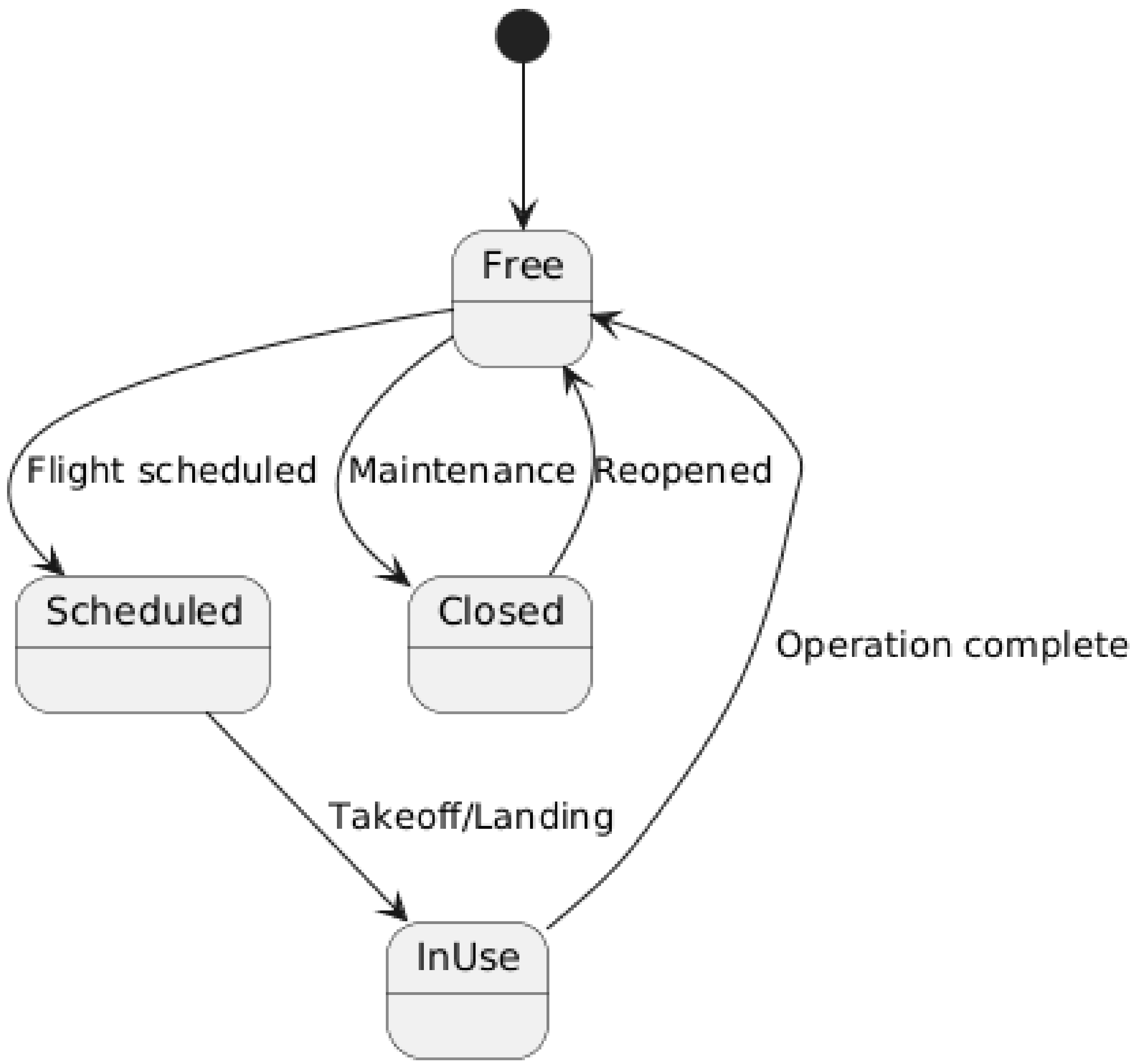
***Objective*** *:*

*The goal is to clearly map the behavioral workflows and object lifecycles to ensure the system design logically fulfills its operational requirements and serves as a foundation for implementation.*

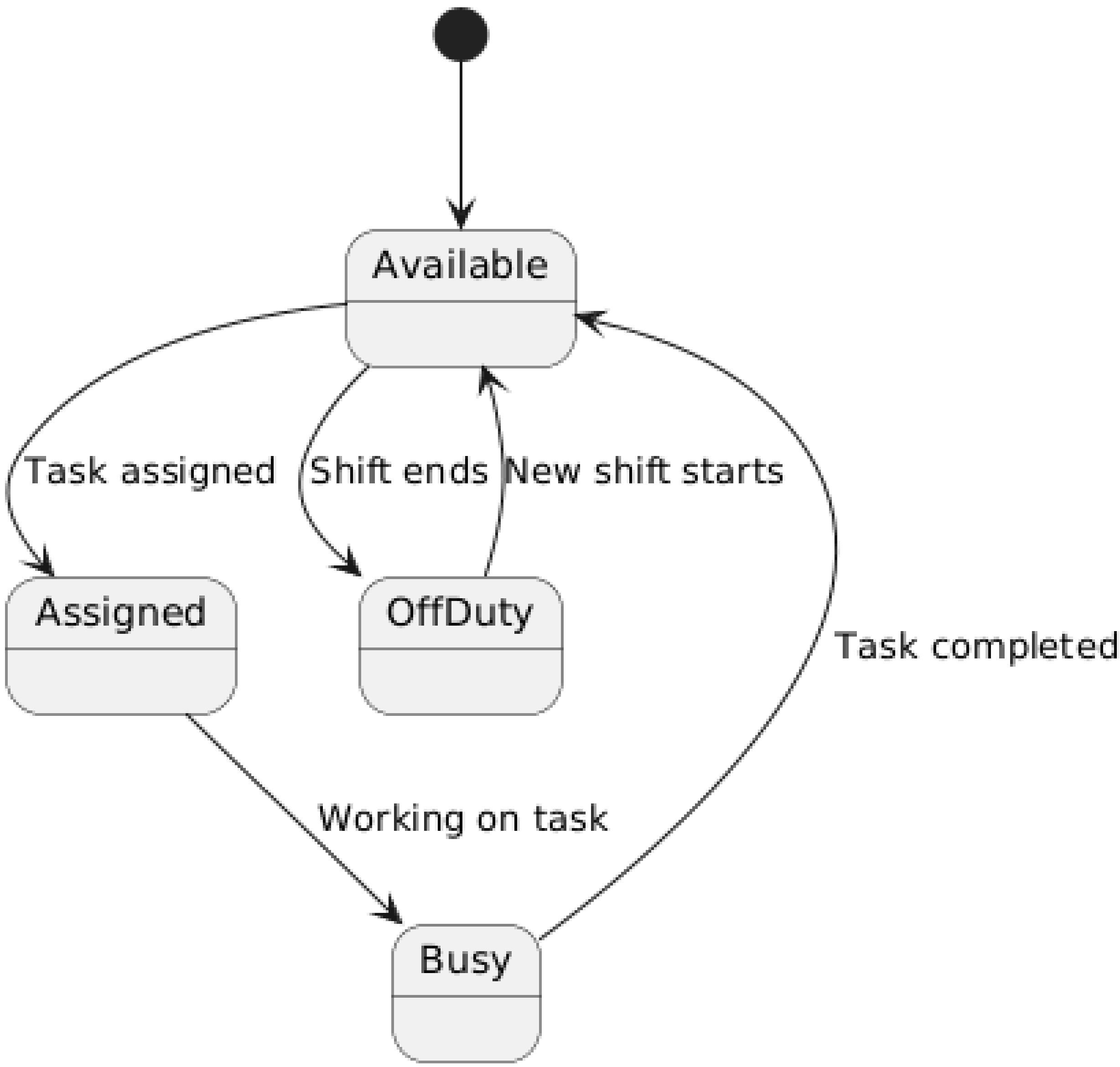
## *UC-ARM-01: Gate Allocation State Diagram*


Available
Flight assigned
Fault detected
Fixed
Reserved
Maintenance
Aircraft departs
Aircraft arrives
Occupied

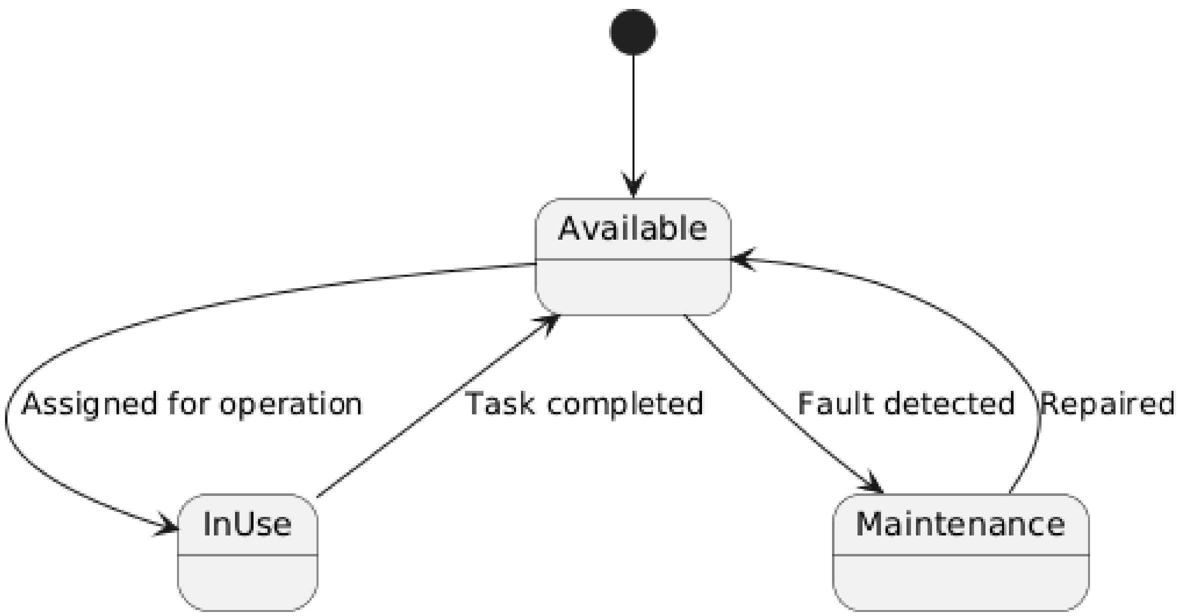
***UC-ARM-02: Runway Scheduling State Diagram***




***UC-ARM-03: Staff Allocation State Diagram***




***UC-ARM-04: Equipment Management State Diagram***




***Airport Management System***

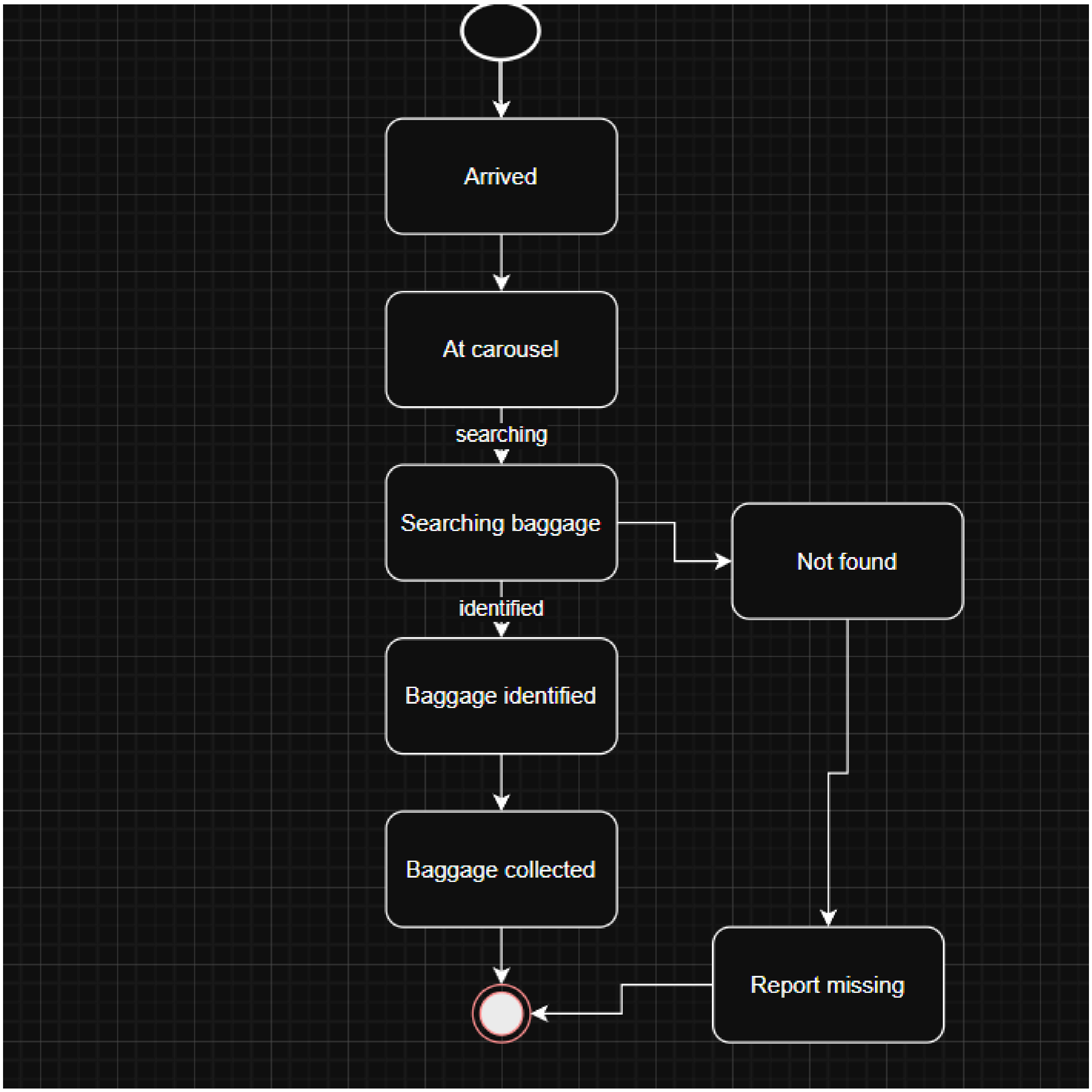
***Prepared by:*** *Albina Blloshmi*

***Course:*** *Software Modeling and Design*

# *UC-08: Claim Baggage at Destination*

*Core Purpose: To model the process of a passenger retrieving baggage upon arrival at the destination airport, ensuring efficient and accurate baggage claim handling.*

*Mapped Requirements: UC-08 (Claim Baggage at Destination). Ensures passengers can locate and collect their baggage or report issues if baggage is missing.*




*• Arrived: The flight has landed and baggage is unloaded.*

*• At Carousel: The passenger is at the baggage claim area.*

*• Searching Baggage: The passenger looks for their baggage.*

*• Baggage Identified: The passenger recognizes their baggage.*

*• Baggage Collected: The passenger retrieves their baggage.*

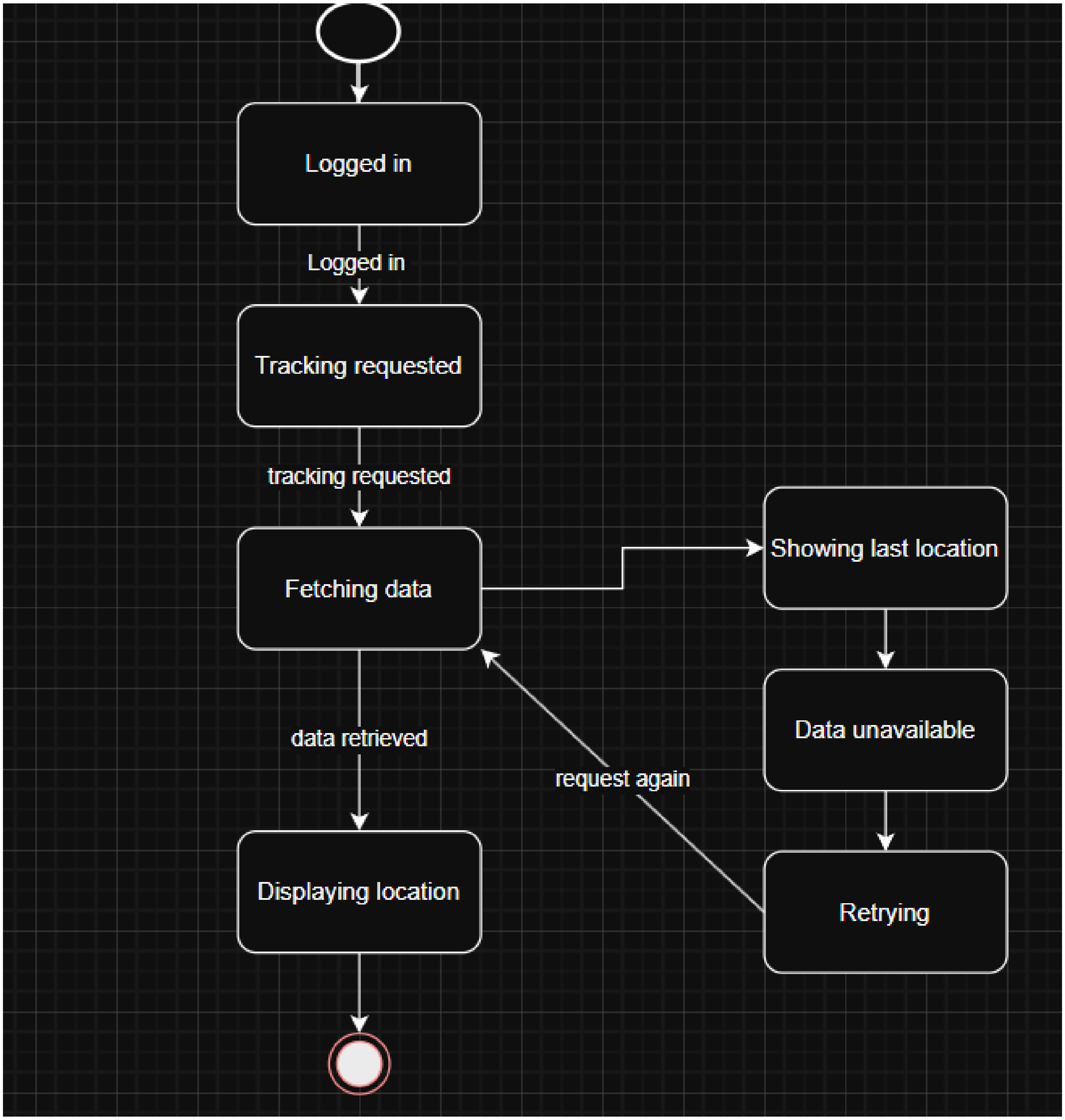
*• Not Found: The passenger cannot locate their baggage.*

*• Report Missing: The passenger reports missing baggage to staff.*

# *UC-09: Track Baggage in Real-Time*

*Core Purpose: To model how passengers track the real-time location of their baggage throughout the journey using the system.*

*Mapped Requirements: UC-09 (Track Baggage in Real-Time). Ensures accurate tracking and visibility of baggage location.*




• *Logged In: The passenger accesses the system.*

• *Tracking Requested: The passenger selects tracking.*

• *Fetching Data: The system retrieves baggage location data.*

• *Displaying Location: The system shows the baggage location.*

• *Data Unavailable: Real-time data cannot be retrieved.*

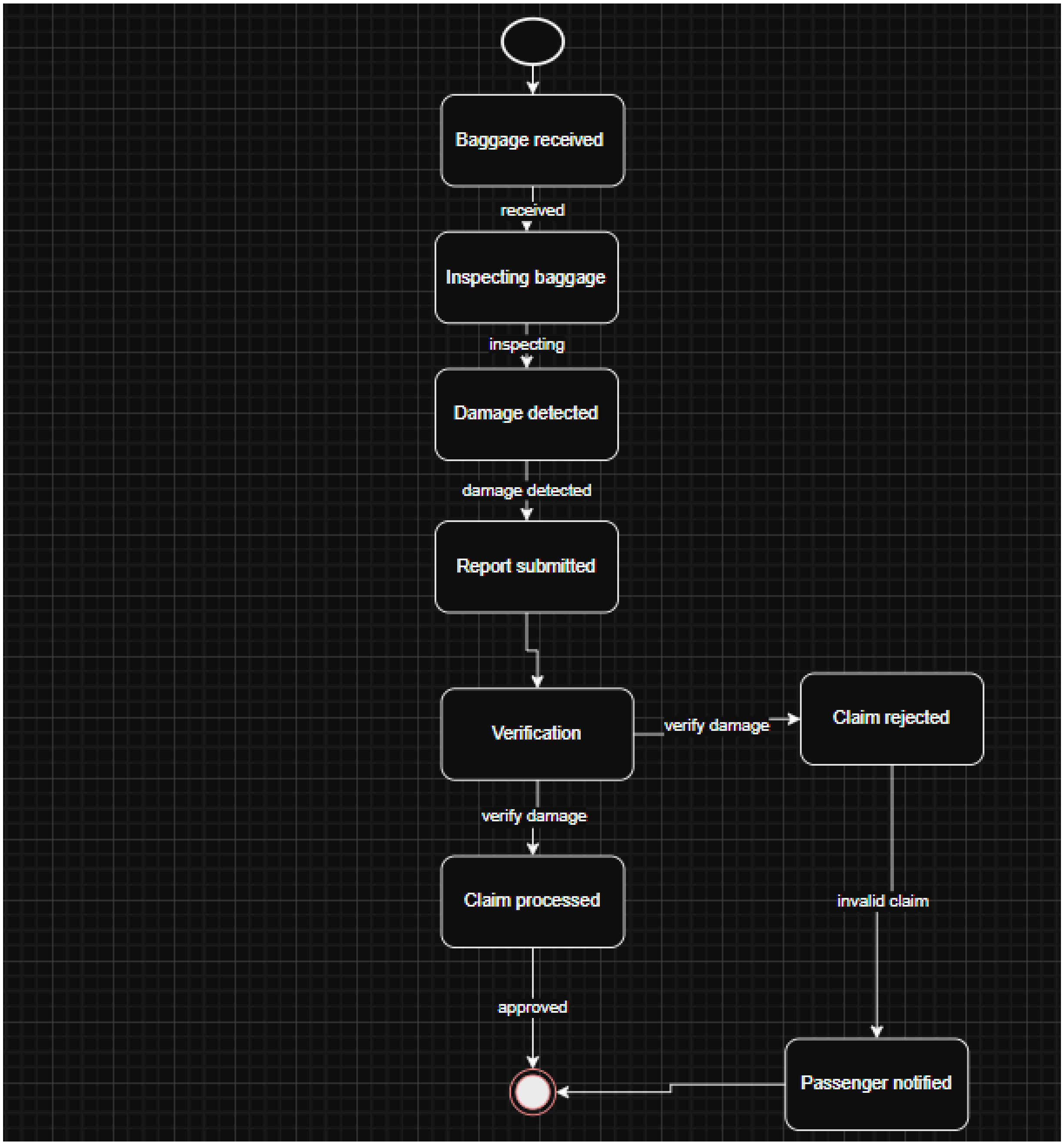
*• Showing Last Location: The system displays last known location.*

*• Retrying: The system attempts to fetch data again.*

# *UC-10: Report Damaged Baggage*

*Core Purpose: To model the process of reporting and handling damaged baggage after arrival, ensuring proper verification and claim processing.*

*Mapped Requirements: UC-10 (Report Damaged Baggage). Ensures damage reports are recorded, verified, and processed efficiently.*




- *Baggage Received: The passenger has received their baggage.*
- *Inspecting Baggage: The passenger checks baggage condition.*
- *Damage Detected: Damage is identified.*
- *Report Submitted: The passenger submits a damage report.*
- *Verification: Staff verifies the reported damage.*

• *Claim Processed: The claim is approved and processed.*

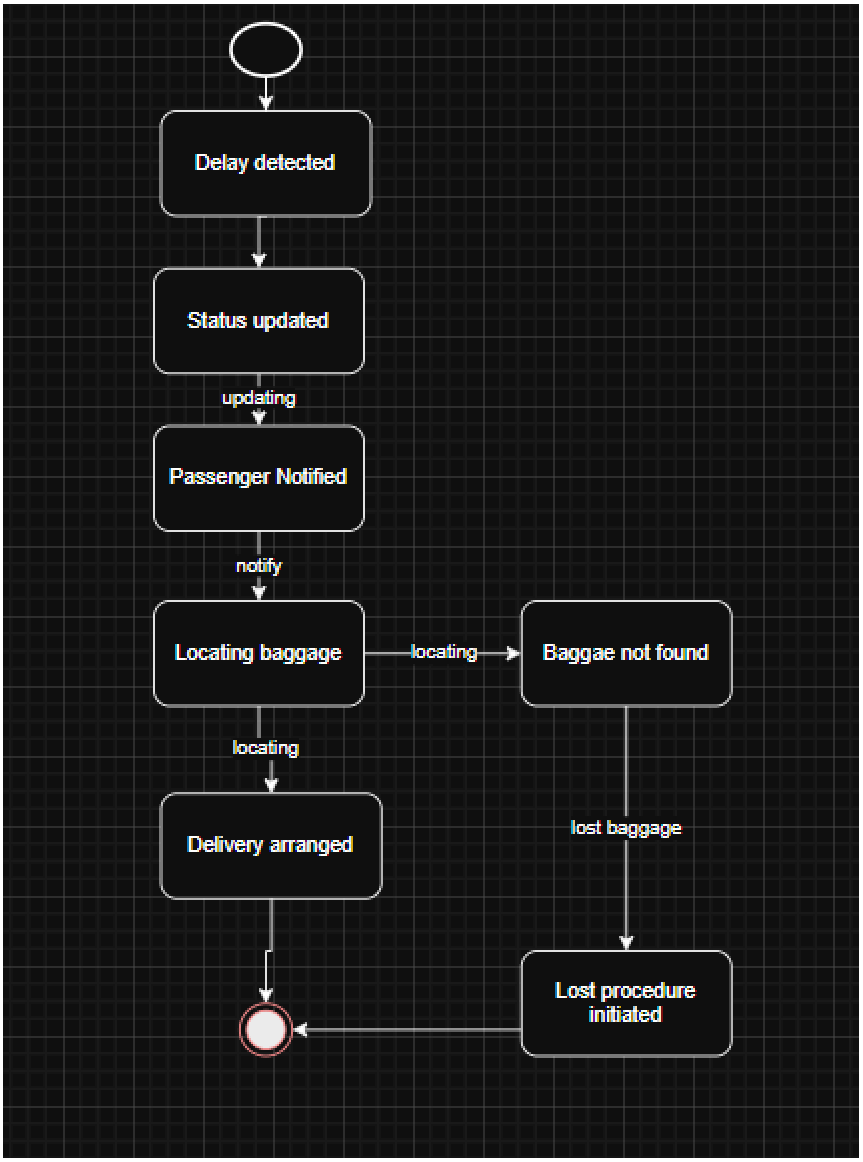
• *Claim Rejected: The claim is denied due to invalid proof.*

• *Passenger Notified: The passenger is informed of the result.*

# *UC-11: Handle Delayed Baggage*

*Core Purpose: To model how the system detects, manages, and resolves delayed baggage cases, ensuring timely updates and resolution.*

*Mapped Requirements: UC-11 (Handle Delayed Baggage). Ensures delayed baggage is tracked, communicated, and resolved efficiently.*

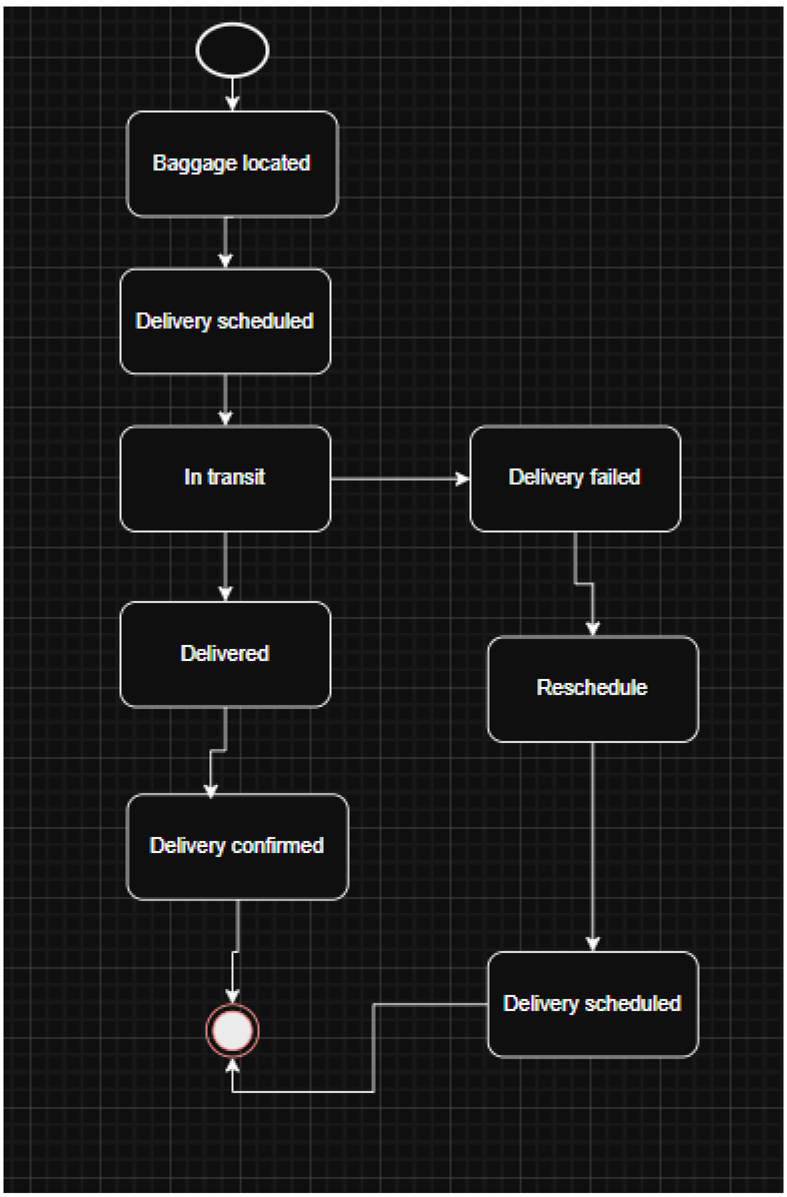


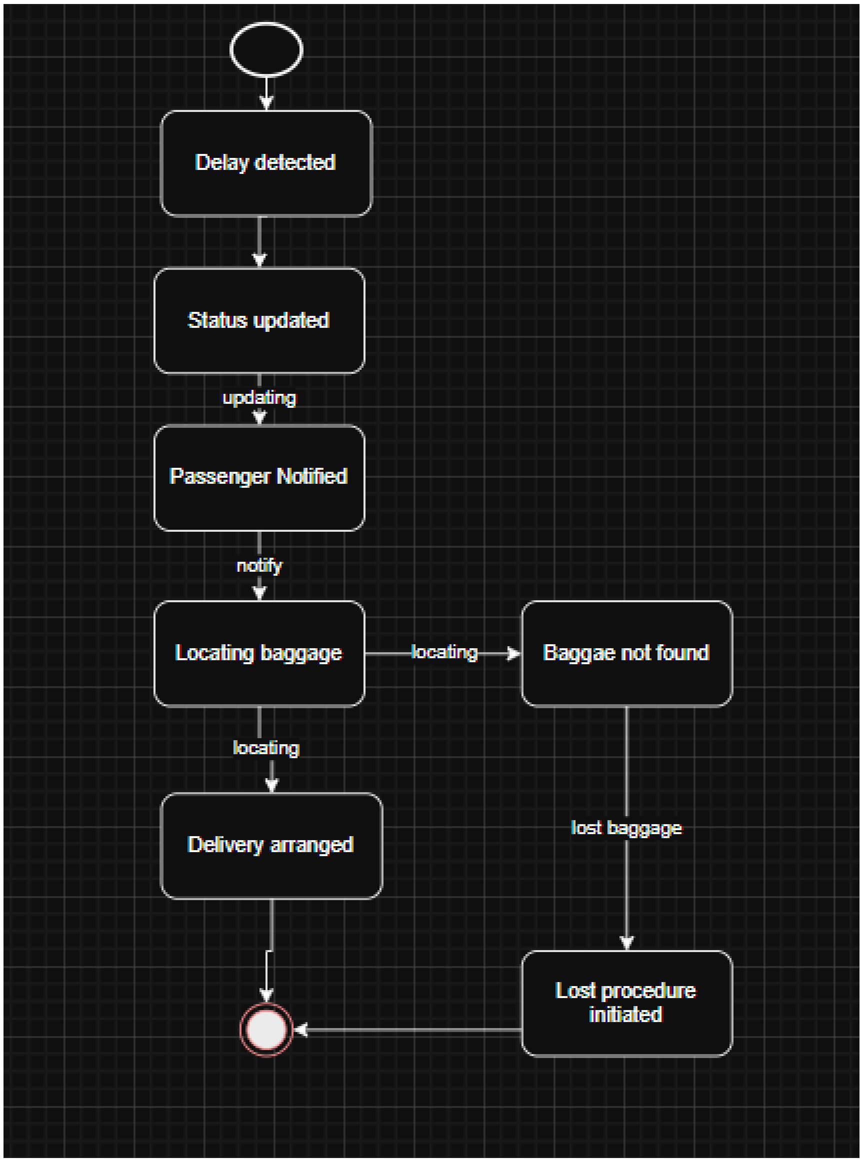

- *Delay Detected: The system identifies baggage delay.*
- *Status Updated: The system updates baggage status.*
- *Passenger Notified: The passenger is informed of the delay.*
- *Locating Baggage: The system attempts to locate baggage.*
- *Delivery Arranged: Delivery is scheduled.*
- *Baggage Not Found: The baggage cannot be located.*
- *Lost Procedure Initiated: Lost baggage handling begins.*

# *UC-12: Deliver Baggage to Passenger*

*Core Purpose: To model the process of delivering baggage to passengers after delay or rerouting, ensuring successful and secure delivery.*

*Mapped Requirements: UC-12 (Deliver Baggage to Passenger). Ensures baggage is delivered efficiently and receipt is confirmed.*

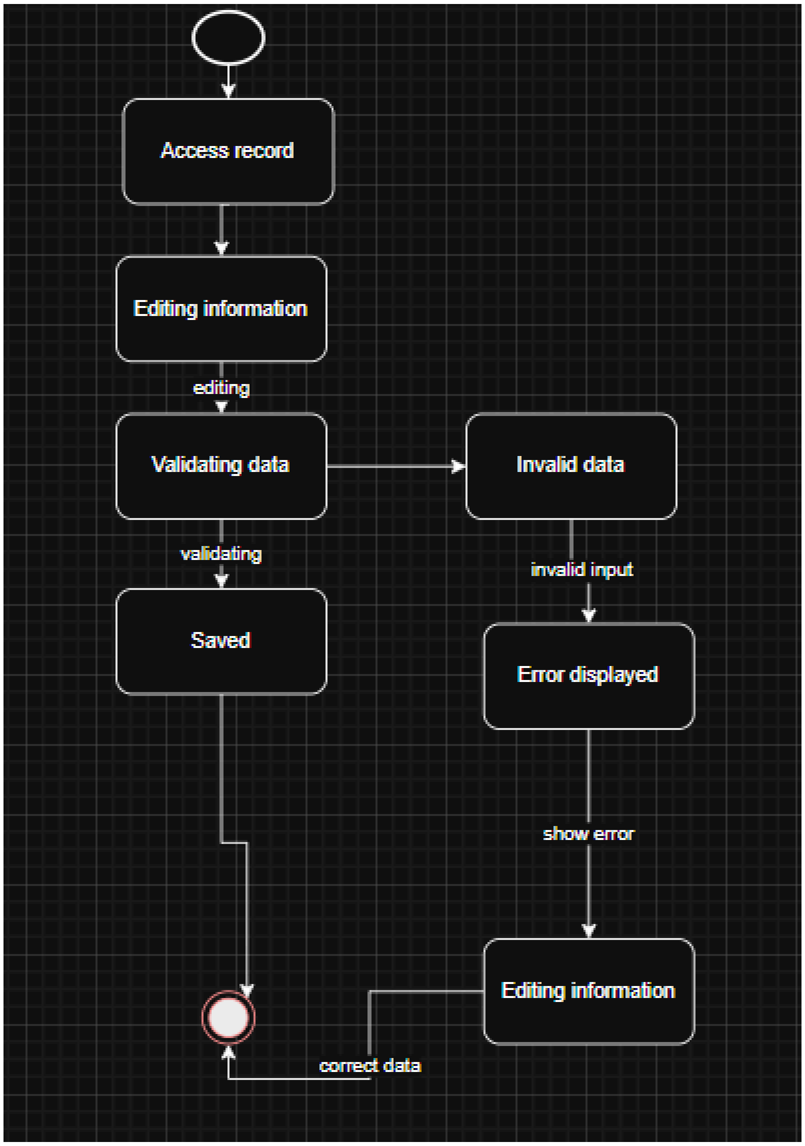





# *UC-13: Update Baggage Information*

*Core Purpose: To model how staff update baggage information in the system, ensuring data accuracy and system consistency.*

*Mapped Requirements: UC-13 (Update Baggage Information). Ensures baggage data is properly maintained and updated.*




- *Access Record: Staff opens baggage record.*
- *Editing Information: Staff updates data.*
- *Validating Data: The system checks data accuracy.*
- *Saved: Changes are successfully stored.*
- *Invalid Data: Incorrect input is detected.*
- *Error Displayed: The system shows an error message.*

## *Airport Management System*

***Prepared by:*** *Eneida Balliu*

***Course:*** *Software Modeling and Design*

### *Document Overview:*

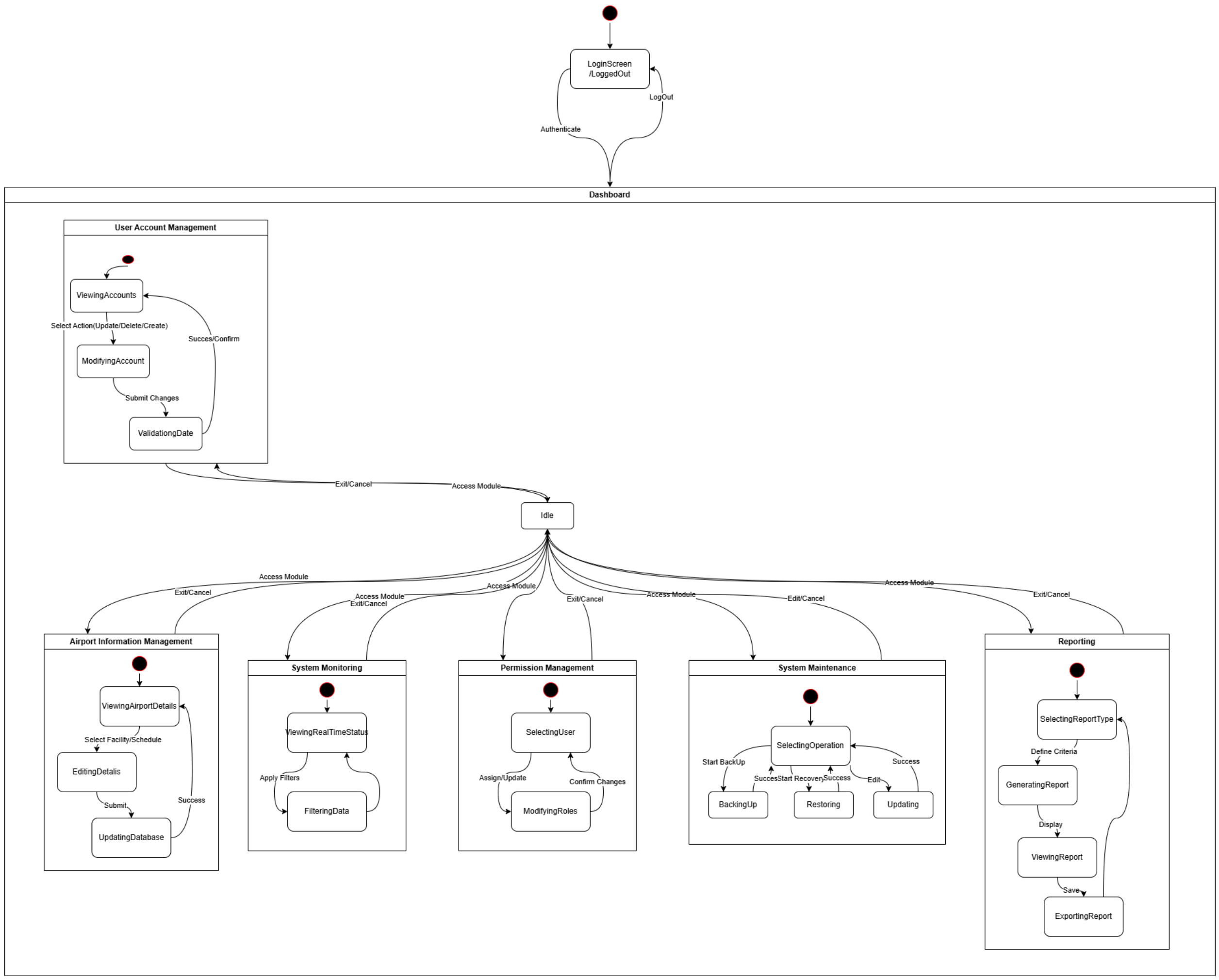
*This document presents the state machine models for the Airport Management System (AMS). It illustrates the exact states, events, and transitions of the system's main objects.*

### *Objective :*

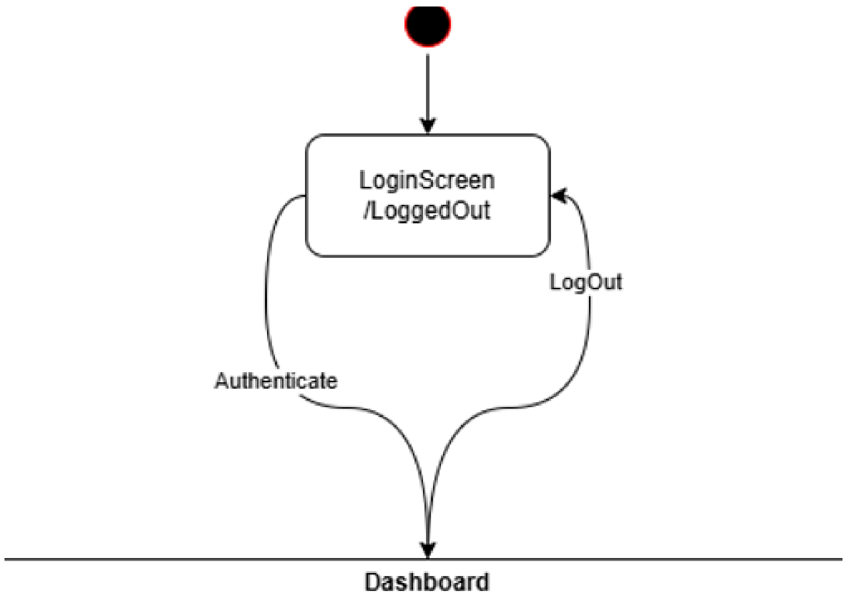
*The goal is to clearly map the behavioral workflows and object lifecycles to ensure the system design logically fulfills its operational requirements and serves as a foundation for implementation.*

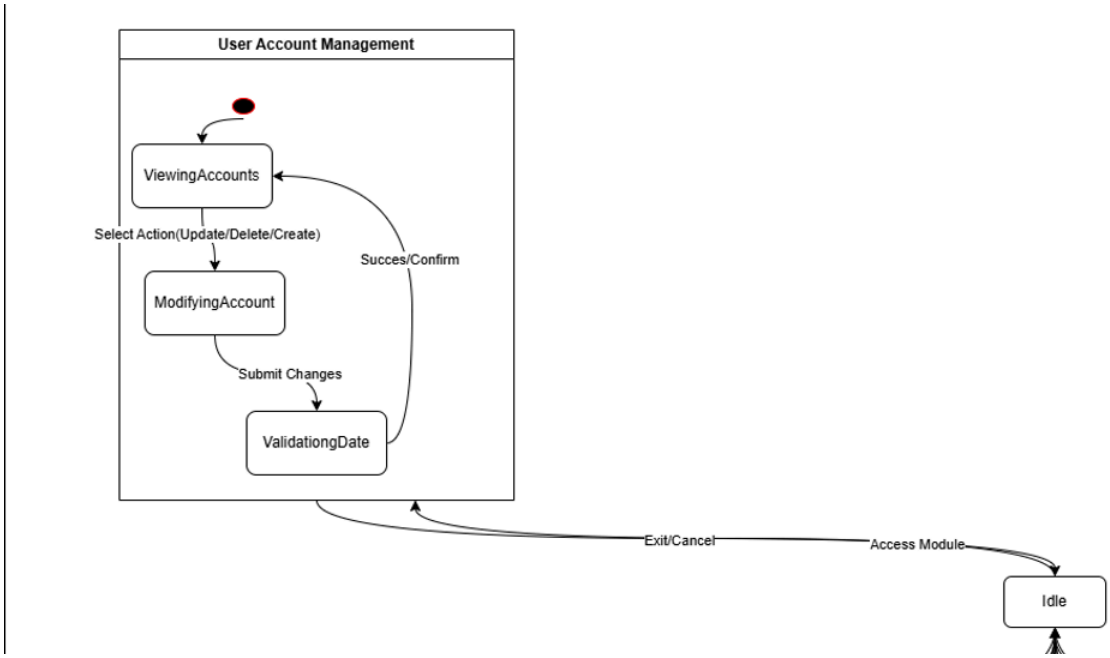
## *State Machine Diagrams*

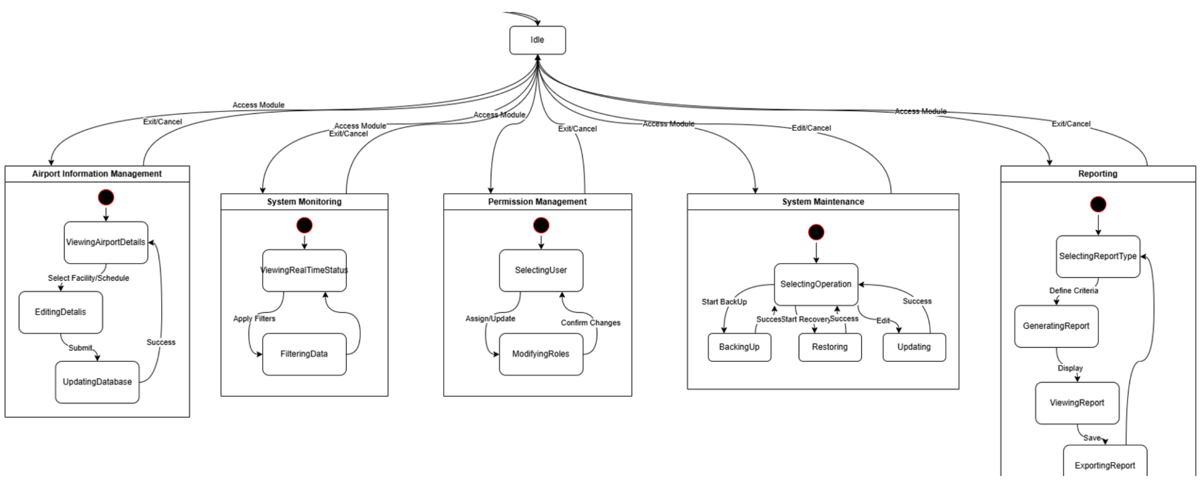
*This section defines the dynamic behavior and state transitions for the core system objects managed by System Administrator.*



LoginScreen
/LoggedOut
Authenticate
LogOut
Dashboard
User Account Management
ViewingAccounts
Select Action(Update/Delete/Create)
ModifyingAccount
Succes/Confirm
Submit Changes
ValidationgDate
Exit/Cancel
Access Module
Idle
Access Module
Exit/Cancel
Access Module
Exit/Cancel
Access Module
Exit/Cancel
Access Module
Edit/Cancel
Access Module
Exit/Cancel
Airport Information Management
ViewingAirportDetails
Select Facility/Schedule
EditingDetalis
Submit
UpdatingDatabase
Success
System Monitoring
ViewingRealTimeStatus
Apply Filters
FilteringData
Permission Management
SelectingUser
Assign/Update
Confirm Changes
ModifyingRoles
System Maintenance
SelectingOperation
Start BackUp
Succes
Start Recovery
Success
Edit
Success
BackingUp
Restoring
Updating
Reporting
SelectingReportType
Define Criteria
GeneratingReport
Display
ViewingReport
Save
ExportingReport


LoginScreen
/LoggedOut
LogOut
Authenticate
Dashboard




This state machine diagram models the dynamic behavior of the Airport Management System from the perspective of the System Administrator, covering the administrative modules defined in the project requirements.

*Functional Logic:*

1. Authentication & Session Control: The system begins in the LoggedOut state. Access to the Dashboard (the main composite state) is only granted after a successful authenticate event. This ensures the security precondition is met for all administrative tasks.
2. The Idle State (Main Menu): Once inside the Dashboard, the system resides in the Idle state. This acts as a central hub where the administrator chooses which module to access. Every module includes an Exit/Cancel transition that returns the system to this Idle state without saving changes, as per the alternative sequences.

3. Module Sub-States:
    a. User & Permission Management (UC_40, UC_43): These states handle the lifecycle of user accounts and access rights. They include a ValidatingData state to ensure information accuracy before the database is updated.
    b. Airport Info Management (UC_41): This state allows for the modification of airport facilities and schedules, moving from an editing phase to a database update phase.
    c. Monitoring & Reporting (UC_42, UC_45): These states focus on data retrieval. The system transitions through filtering, generating, and finally Exporting reports in formats like PDF or CSV.
    d. *System Maintenance (UC_44):* This complex state manages critical operations including Data Backup, Data Recovery, and Configuration Updates.
4. Error Handling (Alternative Flows): The diagram accounts for "Alternative Sequences" described in the documentation, such as invalid data entries or database failures. In these cases, the system transitions to an error state or requests correction instead of confirming the operation.
5. Post-conditions: Successful transitions out of a functional state result in the database being updated and the activity being recorded in the System Logs for auditing purposes.

## *Airport Management System*

***Prepared by:*** *Andri Mucllari*

***Course:*** *Software Modeling and Design*

### *Document Overview:*

*This document presents the state machine diagrams for the Airport Management System (AMS). It shows how key system components behave by outlining their states and the transitions between them.*

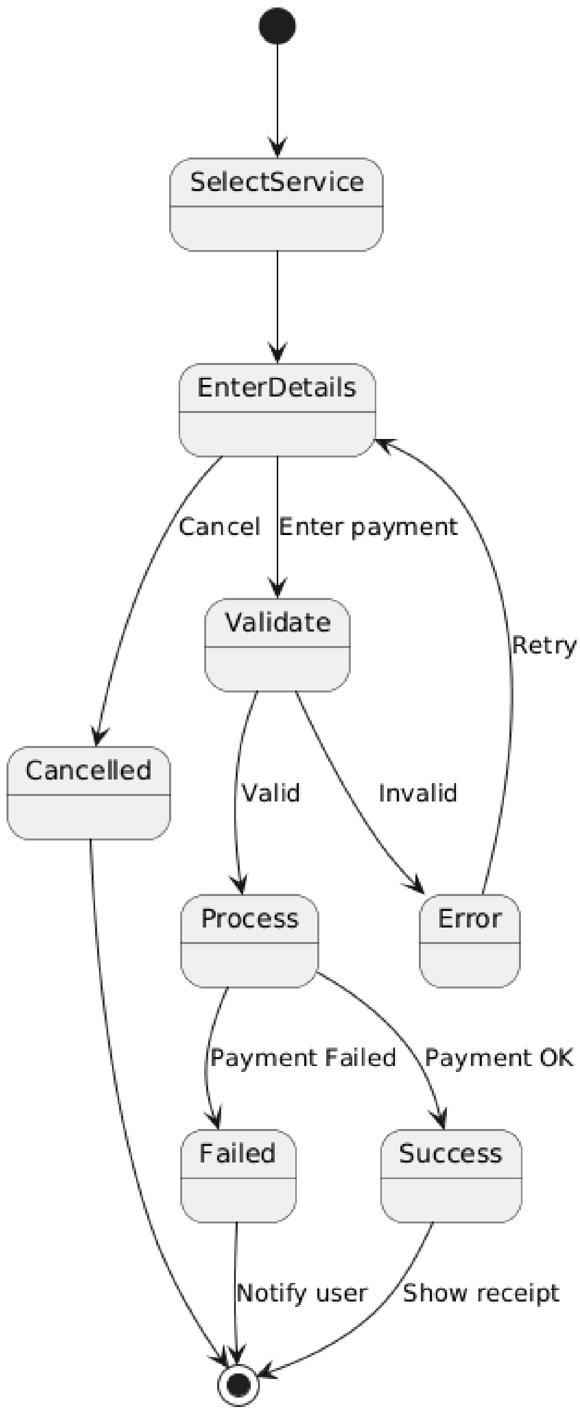
### *Objective :*

*The objective of this document is to represent the system's behavior in a clear and structured way. It focuses on modeling workflows and object lifecycles to ensure the system operates correctly and meets its functional requirement.*

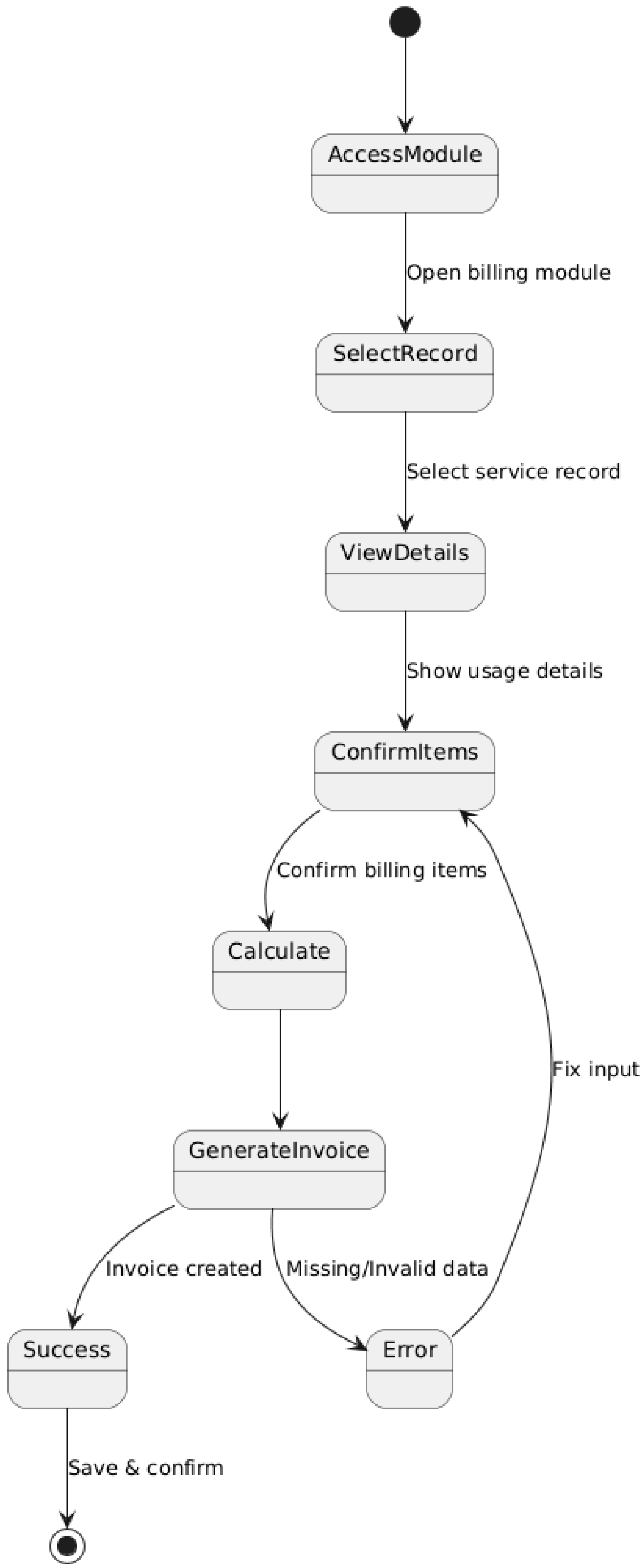
## *State Machine Diagrams*

*This section describes the dynamic behavior of the main system entities by illustrating their possible states and transitions. The diagrams focus on core processes within the financial and service management modules of the system.*

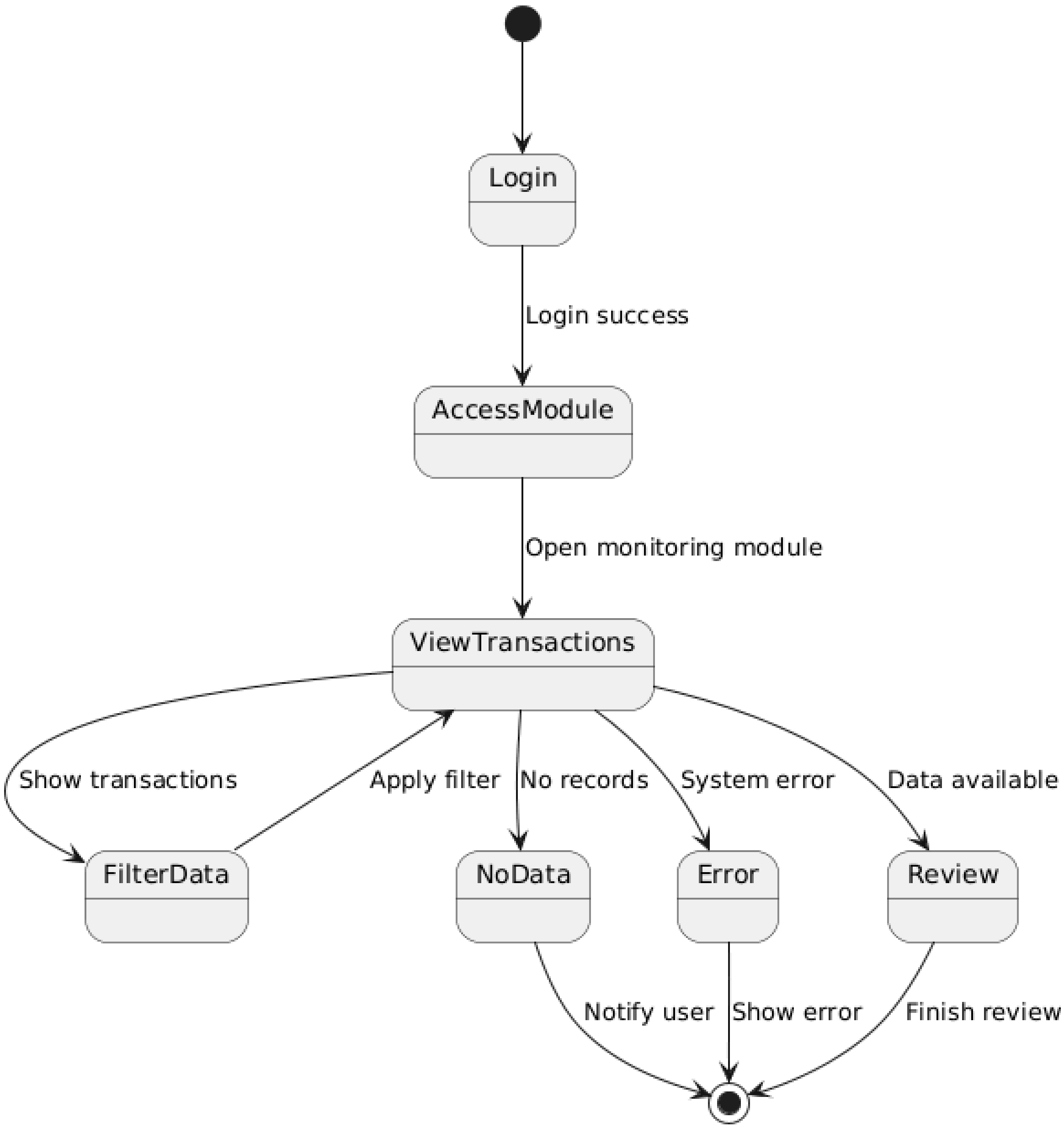
## Processing Airport Service Payments (UC_50)



## Generating Airport Service Invoices (UC_51)



**Monitoring Financial Transactions (UC_52)**



## *Airport Management System*

***Prepared by:*** *Alvi Elezi*

***Course:***
*Software Modeling and Design*

***Document Overview:***

*This document presents the state machine models for the Airport Management System (AMS). It illustrates the exact states, events, and transitions of the system's main objects.*

***Objective:***

*The goal is to clearly map the behavioral workflows and object lifecycles to ensure the system design logically fulfills its operational requirements and serves as a foundation for implementation.*

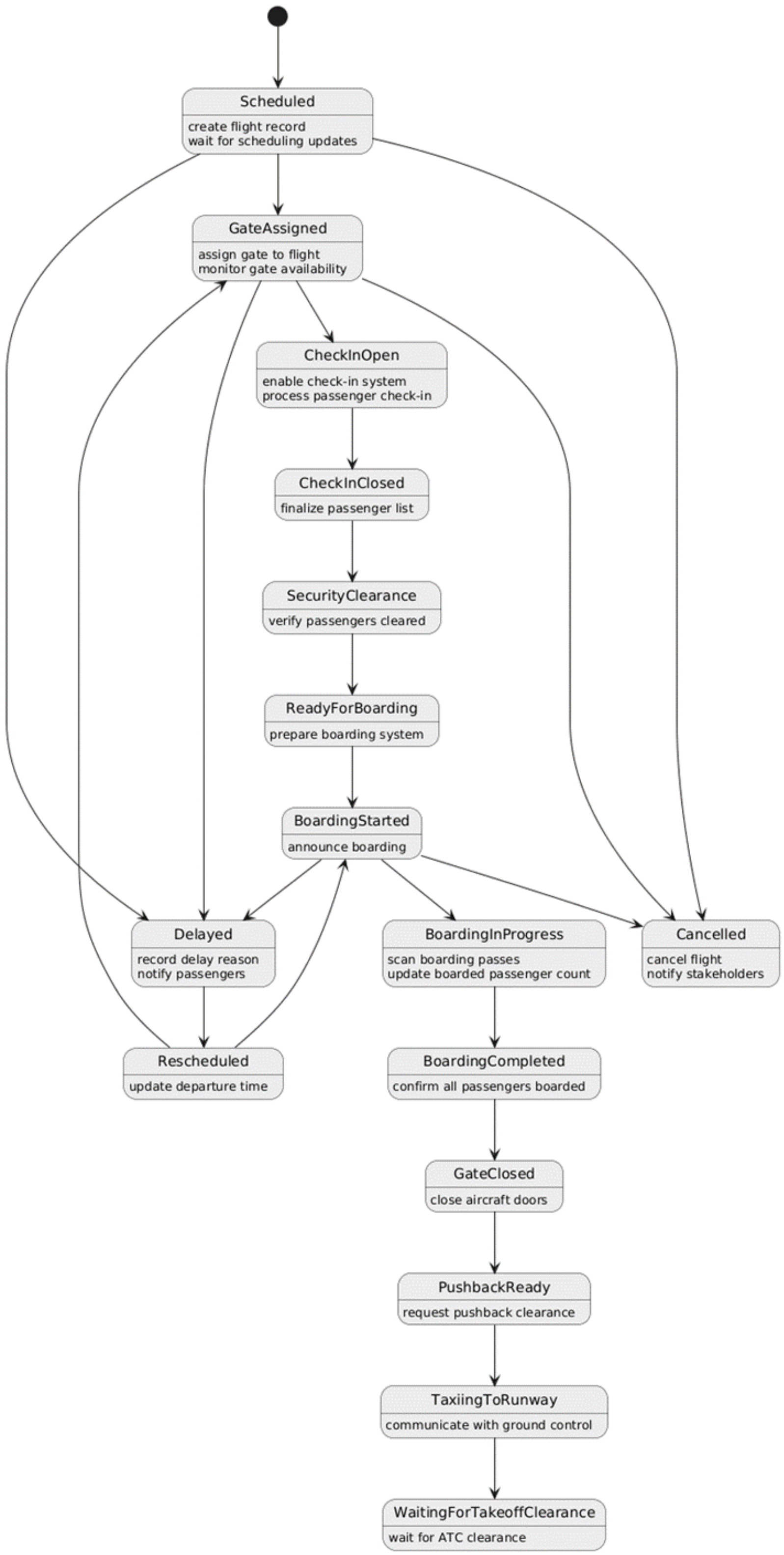
## *State Machine Diagrams*

*This section defines the dynamic behavior and state transitions for the core system objects managed by the Flight Operations module.*

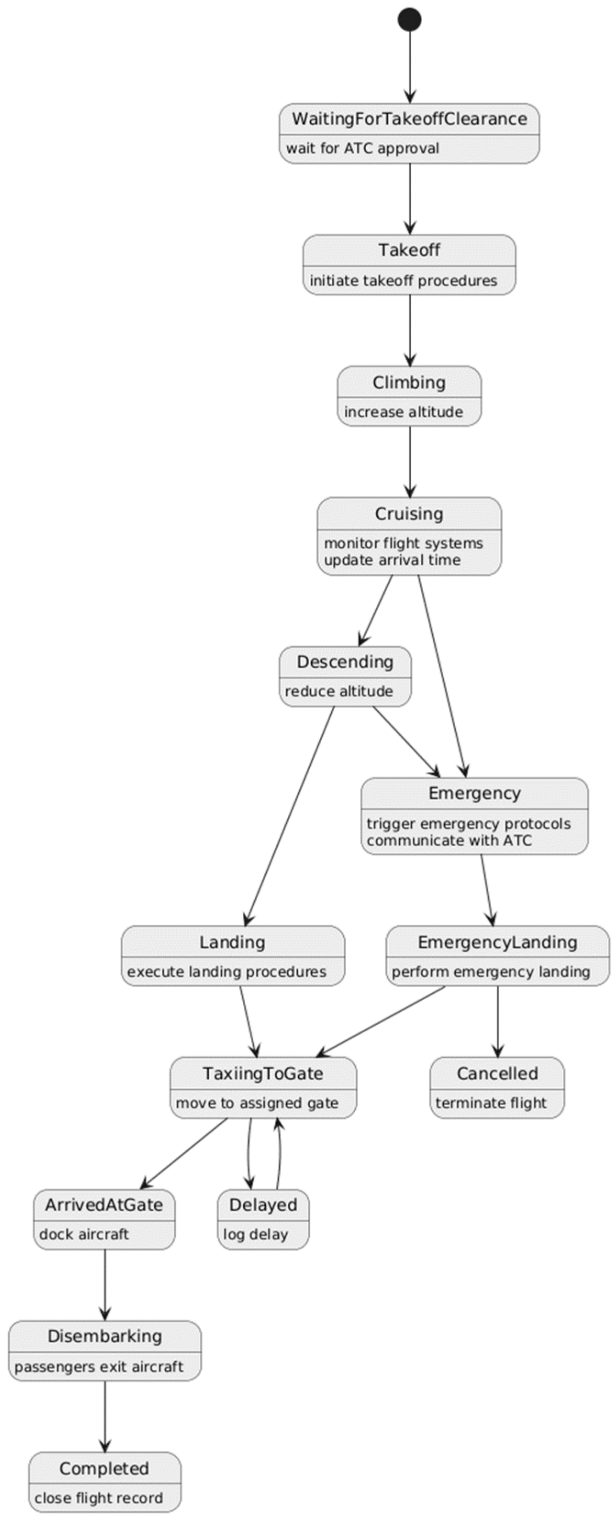
## *How the module is divided:*

The Flight Operations state model is divided into two diagrams to improve readability and maintainability. The first diagram represents pre-flight and ground operations, including scheduling, check-in, boarding, and taxiing. The second diagram represents in-flight operations and arrival processes, including takeoff, cruising, landing, and exception handling such as delays and emergencies.

## Pre - Flight operations diagram:



## Flight execution diagram:



## *Airport Management System*

***Prepared by:*** *Ani Velo*

***Course:*** *Software Modeling and Design*

### *Document Overview:*

*This document presents the state machine models for the* ***Security Management System (SMS)*** *within the Airport Management System. It illustrates the exact states, events, and transitions of the system's core security objects—specifically the* ***Access Request*** *and the* ***Security Incident****.*

### *Objective :*

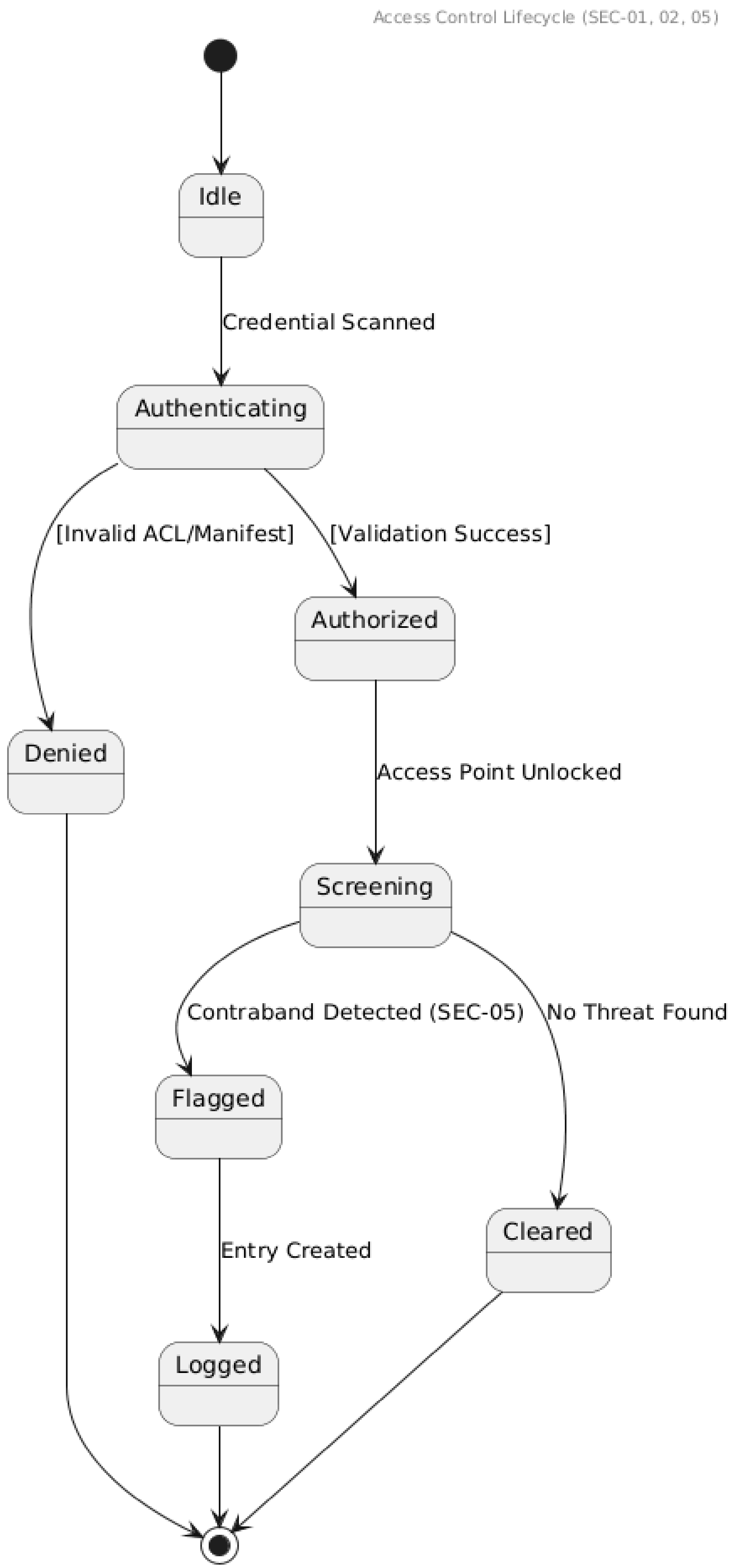
*The goal is to clearly map the behavioral workflows and object lifecycles to ensure the system design logically fulfills its operational requirements. It focuses on ensuring that security protocols, such as identity validation and emergency lockdowns, are executed in a structured, fail-safe manner that serves as a foundation for implementation.*

### *4.1 Access Control Lifecycle (Passenger & Staff Roles)*

- *Core Purpose: To model the exact states an access request passes through, from initial credential scan by a passenger or staff member to final clearance or denial.*
- *Mapped Requirements: Covers **UC-SEC-01** (Passenger Screening), **UC-SEC-02** (Area Access), and **UC-SEC-05** (Prohibited Item Log).*

### *State Descriptions:*

- ***Idle:** The system is in a ready state, waiting for a hardware scan at a checkpoint.*
- ***Authenticating:** The system is actively validating scanned ID/Manifest data against the database.*
- ***Authorized:** Validation is successful; the system has triggered the physical gate to unlock.*
- ***Screening:** A physical search is being performed; the system is waiting for the Officer to input results.*
- ***Flagged:** The system has detected a threat/contraband and has initialized a Seizure Log (UC-SEC-05).*
- ***Cleared/Denied:** Final terminal states where the access attempt is recorded and closed.*


Access Control Lifecycle (SEC-01, 02, 05)
Idle
Credential Scanned
Authenticating
[Invalid ACL/Manifest]
[Validation Success]
Authorized
Denied
Access Point Unlocked
Screening
Contraband Detected (SEC-05)
No Threat Found
Flagged
Entry Created
Cleared
Logged

### *4.2 Security Incident Lifecycle (Officer & Supervisor Roles)*

- *Core Purpose: To define the operational statuses of a security threat to ensure no incident is closed without proper oversight, evidence logging, and 2-Factor Authentication (2FA).*
- *Mapped Requirements: To define the operational statuses of a security threat to ensure no incident is closed without proper oversight, evidence logging, and 2-Factor Authentication (2FA).*

### *State Descriptions:*

- ***Reported:*** *A new incident entry has been initialized in the system (UC-SEC-03).*
- ***Investigating:*** *An Officer ID has been linked to the ticket, and surveillance feeds are being reviewed.*
- ***Escalated:*** *The priority attribute has been set to "Critical," moving the system into high-alert.*
- ***Lockdown Pending:*** *The system is issuing a 2FA challenge to the Supervisor to authorize facility-wide locking.*
- ***Active Lockdown:*** *The system has successfully verified credentials and transmitted "Lock" signals to all hardware (UC-SEC-06).*
- ***Resolved/Archived:*** *The incident is closed with a digital signature and moved to secure 30-day storage.*

## *Airport Management System*

***Prepared by:*** *Betül Kaan*

***Course:*** *Software Modeling and Design*

### *Document Overview:*

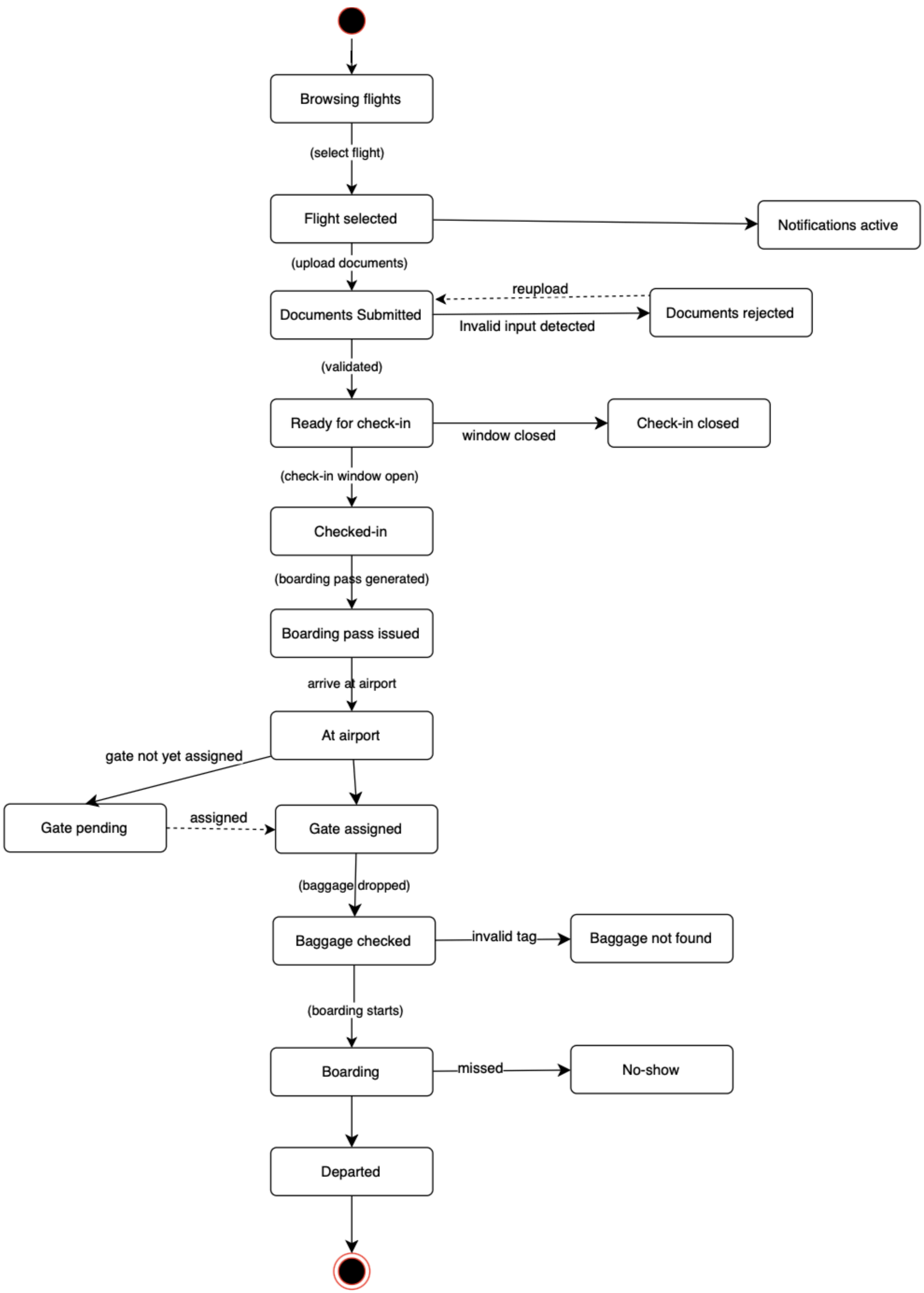
*This document presents the state machine models for the Airport Management System (AMS). It illustrates the exact states, events, and transitions of the system's main objects.*

### *Objective :*

*The goal is to clearly map the behavioral workflows and object lifecycles to ensure the system design logically fulfills its operational requirements and serves as a foundation for implementation.*

## *State Machine Diagrams*

*This section defines the dynamic behavior and state transitions for the core system objects managed within the System Administration models for the Airport Management System.*


Browsing flights
(select flight)
Flight selected
Notifications active
(upload documents)
reupload
Documents Submitted
Invalid input detected
Documents rejected
(validated)
Ready for check-in
window closed
Check-in closed
(check-in window open)
Checked-in
(boarding pass generated)
Boarding pass issued
arrive at airport
At airport
gate not yet assigned
Gate pending
assigned
Gate assigned
(baggage dropped)
Baggage checked
invalid tag
Baggage not found
(boarding starts)
Boarding
missed
No-show
Departed

### *5.1 Passenger Service Lifecycle (Passenger Service Module)*

- ***Core Purpose:*** *To model the exact states a passenger passes through from initial flight browsing to final departure, capturing both the successful journey and all exception flows arising from document validation, check-in timing, baggage handling, and gate assignment.*

***Mapped Requirements:*** *Ensures a passenger can successfully search flights, submit and validate travel documents, complete online check-in, download a boarding pass, track baggage, check gate information, and board their flight — with alternative paths handled at every critical point.*

### *State Descriptions:*

- ***Browsing flights:*** *The passenger has accessed the system and is searching for available flights by date, destination, or flight number (UC-01).*
- ***Flight selected:*** *The passenger has chosen a specific flight and holds a valid booking reference.*
- ***Notifications active:*** *The passenger has opted in to receive real-time alerts for delays, gate changes, and boarding calls. This state runs in parallel and can be entered as soon as a flight is selected (UC-07).*
- ***Documents submitted:*** *The passenger has uploaded required travel documents such as a passport or visa to the system (UC-02).*
- ***Documents rejected:*** *The uploaded file was in an unsupported format or failed validation. The passenger must re-upload corrected documents before proceeding.*
- ***Ready for check-in:*** *Travel documents have been validated and the passenger is eligible to begin online check-in once the check-in window opens.*
- ***Check-in closed:*** *The passenger attempted check-in outside the permitted time window. The passenger is redirected to contact customer support (UC-03 alternative).*
- ***Checked-in:*** *The passenger has successfully completed online check-in and their status is confirmed in the system (UC-03).*
- ***Boarding pass downloaded:*** *The passenger has downloaded their boarding pass as a PDF or image file, ready for use at the airport (UC-04).*
- ***At airport:*** *The passenger has physically arrived at the airport on the day of travel.*
- ***Gate pending:*** *A gate has not yet been assigned to the flight. The passenger is advised to check back or rely on notifications (UC-05 alternative).*
- ***Gate assigned:*** *The departure gate and terminal details are confirmed and visible to the passenger (UC-05).*
- ***Baggage checked:*** *The passenger's luggage has been dropped off and registered in the system by airport staff (UC-06).*
- ***Baggage not found:*** *The baggage tag number entered was invalid or unrecognized. The passenger is prompted to contact support (UC-06 alternative).*
- ***Boarding:*** *The boarding call has been issued and the passenger is in the process of boarding the aircraft.*
- ***No-show:*** *The passenger failed to board before the gate closed, resulting in a terminal exception state.*
- ***Departed:*** *The passenger has successfully boarded and the flight has departed. This is the final successful end state.*

***Airport Management System***

***Prepared by:*** *Eneida Balliu, Betül Kaan, and Ajla Alcani*

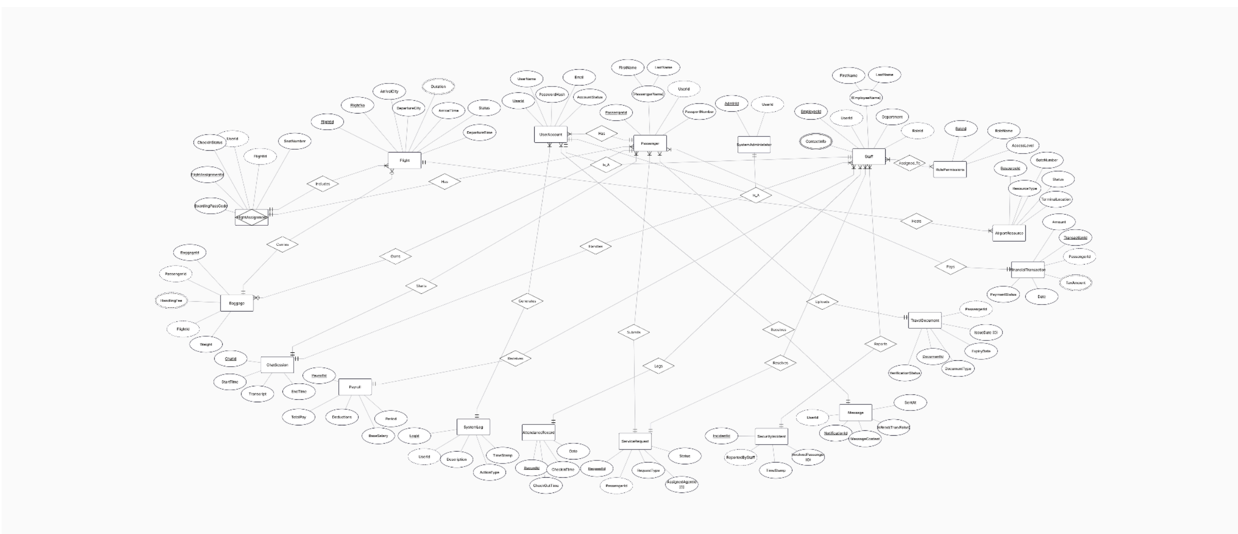
***Course:*** *Software Modeling and Design*

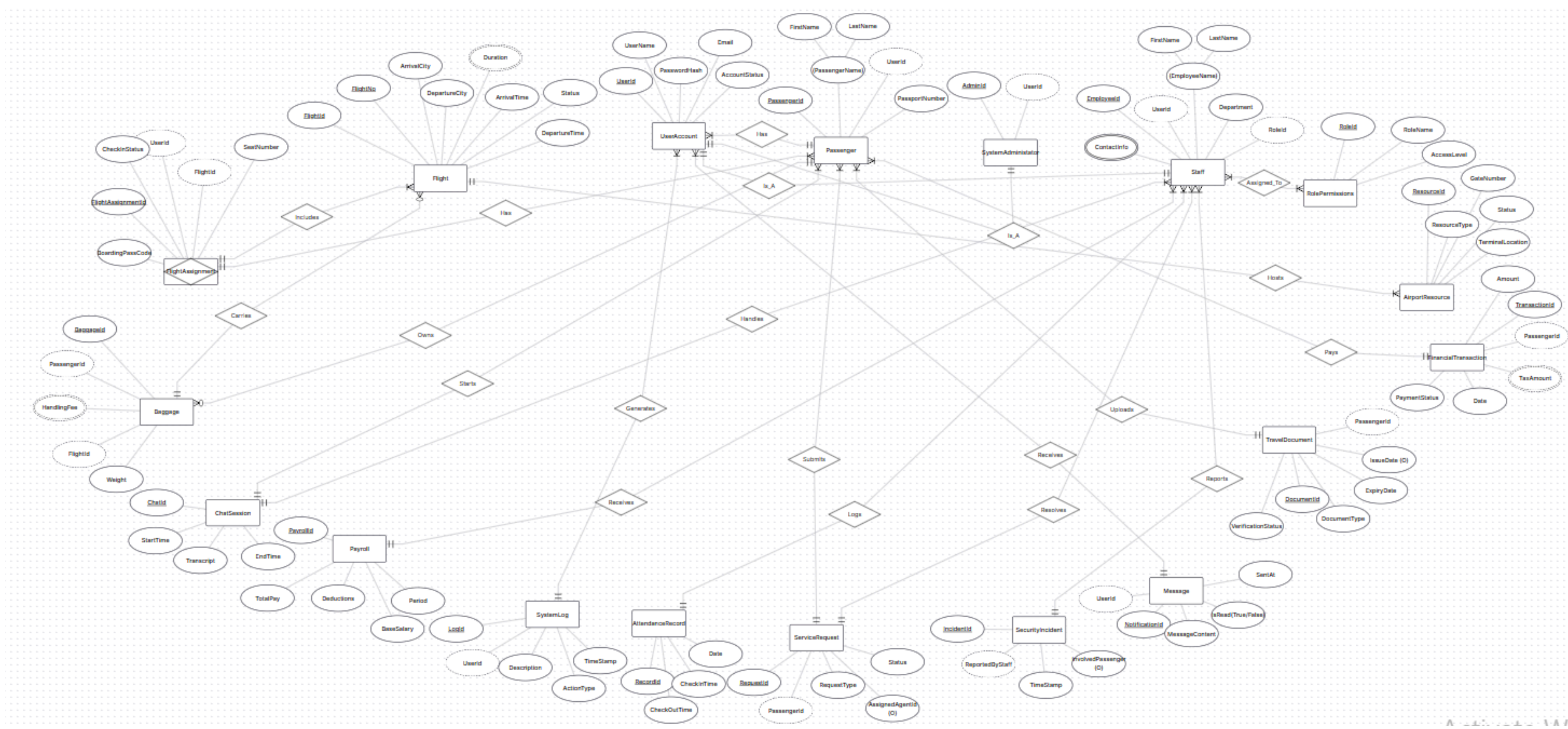
***Document Overview***

*This document presents the state machine models and the comprehensive Entity Relationship Diagram (ERD) for the Airport Management System (AMS). It illustrates the exact states and transitions of the system's main objects, as well as the complete database architecture across all modules.*

***Objective***

*The goal is to clearly map the behavioral workflows and structural data models to ensure the system design logically fulfills its operational requirements and serves as a strong foundation for implementation.*




FlightAssignment
CheckInStatus
UserId
FlightId
SeatNumber
FlightAssignmentId
BoardingPassCode
Flight
FlightId
FlightNo
ArrivalCity
DepartureCity
Duration
ArrivalTime
Status
DepartureTime
Includes
Has
Carries
Baggage
BaggageId
PassengerId
HandlingFee
FlightId
Weight
Owns
Starts
ChatSession
ChatId
StartTime
Transcript
EndTime
Payroll
PayrollId
TotalPay
Deductions
Period
BaseSalary
Receives
Generates
SystemLog
LogId
UserId
Description
TimeStamp
ActionType
AttendanceRecord
Date
RecordId
CheckInTime
CheckOutTime
UserAccount
UserName
Email
UserId
PasswordHash
AccountStatus
Has
Is_A
Handles
Passenger
FirstName
LastName
(PassengerName)
UserId
PassengerId
PassportNumber
Submits
ServiceRequest
RequestId
Status
RequestType
PassengerId
AssignedAgentId (O)
Logs
SystemAdministrator
AdminId
UserId
Is_A
Receives
Resolves
SecurityIncident
IncidentId
ReportedByStaff
TimeStamp
InvolvedPassenger (O)
Message
UserId
NotificationId
MessageContent
IsRead(True/False)
SentAt
Staff
FirstName
LastName
(EmployeeName)
EmployeeId
UserId
Department
RoleId
ContactInfo
Assigned_To
RolePermissions
RoleId
RoleName
AccessLevel
Hosts
AirportResource
ResourceId
ResourceType
GateNumber
Status
TerminalLocation
Pays
FinancialTransaction
Amount
TransactionId
PassengerId
TaxAmount
PaymentStatus
Date
Uploads
TravelDocument
PassengerId
IssueDate (O)
ExpiryDate
DocumentId
DocumentType
VerificationStatus
Reports